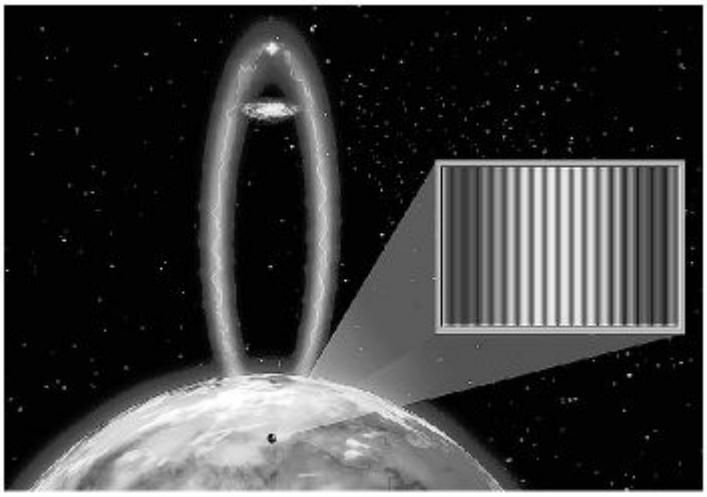


Figure 7.3 Light from a distant quasar, split and focused by an intervening galaxy, will, in principle, yield an interference pattern. If an additional detector, which allows the determination of the path taken by each photon, were switched on, the ensuing photons would no longer fill out an interference pattern.

Has our turning the detector on or off in the twenty-first century had an effect on the motion of photons some billions of years earlier? Certainly not. Quantum mechanics does not deny that the past has happened, and happened fully. Tension arises simply because the concept of *past* according to the quantum is different from the concept of *past* according to classical intuition. Our classical upbringing makes us long to say that a given photon *did* this or *did* that. But in a quantum world, our world, this reasoning imposes upon the photon a reality that is too restrictive. As we have seen, in quantum mechanics the norm is an indeterminate, fuzzy, hybrid reality consisting of many strands, which only crystallizes into a more familiar, definite reality when a suitable observation is carried out. It is not that the photon, billions of years ago, decided to go one way around the galaxy or the other, or both. Instead, for billions of years it has been in the quantum norm—a hybrid of the possibilities.

The act of observation links this unfamiliar quantum reality with everyday classical experience. Observations we make today cause one of the strands of quantum history to gain prominence in our recounting of the past. In this sense, then, although the quantum evolution from the past until now is unaffected by anything we do now, the story we tell of